

Eungchan Kang

Senior Software Engineer

Senior Software Engineer with **5+ years** of professional experience in software architecture, AI engineering, and cloud platforms, designing and operating multi-tenant SaaS systems end to end.

Experienced across the full system lifecycle - from architectural design to production deployment - with strong expertise in:

- Software architecture: layered design, domain / API / data / deployment separation
- AI engineering: LLM agents, MCP, harness and graph engineering
- Multi-tenant SaaS and AWS-based cloud systems
- Database performance: RDBMS internals, execution plans, and query tuning

Designed production-scale systems supporting thousands of organizations and large transactional workloads, with measurable performance improvements

Contact

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Portfolio
yeschan119.com

Skills

Language : C, C++, C#, Python, Typescript, SQL, PL/SQL

AI Engineering : LLM Agents, MCP, Harness Engineering, Graph Engineering

Database : Oracle, MySQL, MsSQL, Tiberio

Cloud | Infrastructure : AWS, Azure, Git, Docker

Framework | Platform : Angular, ASP.Net Core, Flutter

Professional Experience

2026.05 - Current

Senior Software Engineer

Ko&Clo, Inc, Seoul South Korea

Key Responsibilities

- Designed and automated core business workflows, including sales, inventory, ordering, and settlement, as a unified ERP system.
- Applied software architecture principles to the ERP system, designing and implementing the domain, API, data, and deployment layers.
- Built an AI agent-based environment that connects system queries, controls, and automation tasks so operators can control the software more directly.

2024.02 - 2026.04

Full Stack Software Engineer

Kokomo Solutions, Inc, Chicago IL United States

Key Responsibilities

- Designed and implemented scalable backend systems using ASP.NET Core MVC
- Developed Angular-based front-end features including kiosks, chat services, and analytics dashboards
- Built REST APIs supporting real-time incident reporting and classification
- Collaborated with cross-functional teams to deliver production-ready features

Data Platform & Analytics Ownership

- Designed and implemented end-to-end analytics dashboard
- Create/Update/Tune Views/Indexes for high-frequency queries.
- Applied composite indexes across frequently filtered columns
- Applied various partitioning strategies by table workload type
- Reduced dashboard latency by 30%
- Improved query performance by 40%

Performance & Database Engineering

- Optimized MySQL and MSSQL queries under production load
- Improved overall application performance by 30%.
- Implemented database monitoring features for performance management.
- Introduced data isolation techniques to enhance multi-tenant security
- Used various skills (parallel api calls, async) to reduce client-side latency.

2022.01 - 2024.02

Database Engineer (*RDBMS Engine Development*)

Tmaxgroup, Seongnam South Korea

Core member of RDBMS engine team focusing on Optimizer and Statistics modules.

Engine Development

- Designed and implemented statistics collection modules for database indexes
- Identified clustering factor inaccuracies causing performance degradation
- Developed dedicated index statistics collection node
- Achieved 90%+ statistics accuracy improvement
- Improved index scan performance stability

Query Optimization & Plan Management

- Implemented SQL Tuning Advisor features
- Implemented SQL Plan Management (SPM)
- Developed optimizer logic in C++
- Enhanced Join selectivity calculation (5s → 1s improvement)
- Optimized OR-condition handling (3s → 0.8s improvement)
- Reduced targeted feature execution time (827ms → 27ms)

Client Support & Performance Tuning

- Provided SQL tuning support to enterprise clients including banks
- Analyzed execution plans and resolved performance bottlenecks
- Optimized stored procedures and indexing strategies

Legacy Code Modernization

- Migrated C-based legacy components to C++
- Refactored optimizer modules for maintainability and extensibility

2021.09 - 2021.12

Artificial Intelligence Engineer

AIWorkx, Seoul South Korea

Developed AI vision software

- Implemented AI vision solutions using image segmentation and annotation algorithms.
- Developed edge device AI software leveraging object detection algorithms, such as YOLO.

Conducted research and presented findings on AI vision

- Regularly studied and presented research papers on AI vision to identify optimized algorithms and improve implementation strategies.

Education

Bachelor of Science: Information System

Hanyang University - Seoul South Korea

minor : Computer Science

- Total GPA : 4.18 / 4.5
- Major GPA : 4.32 / 4.5